CONTROL-M / JP1 PLAYBOOK

Architecture & Control-M

Enterprise Schedulers & Pipelines

1. 4-Step Pipeline Execution Flow

Every batch execution triggered via 'python3 main.py --run-batch' follows an audited 4-step pipeline designed to prevent unmonitored production drift:

* **STEP 1: Unix Housekeeping: Captures real Linux** OS metrics (df, free, uptime, ps aux), purges data/spool_tmp, verifies daemons.
* **STEP 2: DAG Batch Engine: Resolves schedule** dependencies, executes isolated subprocesses, applies retry loops, and evaluates SLAs.
* **STEP 3: ServiceNow ITIL Triage: Applies Rules** A, B, and C to create P1, P2, and P3 tickets with parent-child correlation in data/servicenow_outbox/.
* **STEP 4: Shift Handover Reports: Generates CSV** check-sheet and dark-themed HTML shift handover dashboard in reports/.

2. Control-M / JP1 Enterprise Mapping

Use this side-by-side mapping when interviewing at MUFG to prove you understand their enterprise tooling:

1. Schedule Definition

BatchGuard:	config/batch_schedule.json
BMC Control-M:	Job Definition XML / Smart Folder
Hitachi JP1:	Jobnet Definition / Group
CA AutoSys:	JIL (Job Instruction Language)
Function:	Centralized schedule & agent configs

2. Dependency Ordering

BatchGuard:	depends_on: ['JOB_01_FX_INGEST']
BMC Control-M:	IN-Conditions & OUT-Conditions
Hitachi JP1:	Preceding Job Execution Conditions
CA AutoSys:	condition: success(JOB_01)
Function:	Prevents downstream start until OK

3. Cyclic Retries

BatchGuard:	max_retries: 2, retry_delay_sec: 2
BMC Control-M:	MAXRERUN=2, RERUNMEM=2M
Hitachi JP1:	Auto-Retry Settings / Interval
CA AutoSys:	n_retrys: 2, retry_interval: 2
Function:	Absorbs transient lock/network spikes

4. SLA Tracking	
BatchGuard:	sla_max_duration_sec: 1.5
BMC Control-M:	Batch Impact Manager (BIM) Alert
Hitachi JP1:	JP1/AJS3 SLA Monitoring
CA AutoSys:	term_run_time / alarm_if_fail
Function:	Alerts when critical path is delayed

5. Alert Forwarding	
BatchGuard:	src/incident_manager.py
BMC Control-M:	Control-M Shout to ServiceNow
Hitachi JP1:	JP1/IM Incident Bridge Forwarder
CA AutoSys:	AutoSys Alarm to SNOW API Bridge
Function:	Converts exit codes to ITIL tickets

3. Operational Actions & Resuming Batches

INTERVIEW SCENARIO: RESUMING FROM FAILURE

If JOB_03 is fixed by the DBA team at 3:15 AM, how do you resume without re-running JOB_01 and JOB_02? In Control-M, you perform a 'Rerun from Failed Job'. In BatchGuard, because states are saved in SQLite, we inspect job_runs, verify JOB_01 and JOB_02 are SUCCESS, and start directly at JOB_03, unblocking JOB_04 upon completion.

* **Force OK: Manually sets job state to Success** in Control-M to allow downstream jobs to proceed when safe.
* **Bypass / Skip: Temporarily disables a job** without breaking DAG topological dependencies.
* **Kill / Terminate: Sends SIGKILL to a hung** batch process exceeding max run time.